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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-2026)
Faculty Name	Ms. G. Mamatha
Student Name	V.VedaPriya
Roll Number	2520030376
Branch	CSE
Course Outcome	CO3-Breadth First Search ,Depth First Search , Minimum Spanning Tree

Case Study Topic: Student Course Connectivity and Academic Network Optimization using Graph Algorithms in EduTrack System

Work Done Summary:

In this case study, students, courses, and academic departments are represented as a graph to analyze connectivity and optimize academic resource management. BFS and DFS algorithms are used to traverse student-course networks and identify connected academic groups, while the Minimum Spanning Tree (MST) algorithm is implemented to design a minimum-cost academic communication and resource-sharing network between departments.

Implementation Details:

1. Students, courses, and departments are modeled as graph vertices.
2. Connections between students and courses are represented as graph edges.
3. BFS (Breadth First Search) is used to analyze course connectivity and identify reachable academic resources for students.
4. DFS (Depth First Search) is used to traverse academic networks and detect connected student groups and department relationships.
5. Minimum Spanning Tree (MST) is implemented to establish the minimum-cost academic communication and resource-sharing network among departments.
6. Adjacency List representation is used for efficient graph storage and faster traversal operations.

7. The system helps optimize student-course allocation, faculty coordination, and academic resource sharing.
8. Graph traversal techniques improve academic planning and connectivity analysis between departments and students.
9. The system supports efficient management of large academic networks with multiple courses and departments.
10. Optimized graph algorithms reduce communication cost and improve overall educational management efficiency.

Result : Screenshots / Output

```
Departments:
0 -> Computer Science
1 -> AI & DS
2 -> ECE
3 -> Mechanical
4 -> Civil

Final MST (Academic Communication Network)

Edge    Weight
0 - 1   2
1 - 2   3
0 - 3   6
1 - 4   5
PS C:\Project>
```

Time Complexity :

1. Graph Representation: **O(V²)**
2. Minimum Spanning Tree Construction: **O(V²)**
3. Edge Selection Process: **O(V²)**
4. Space Complexity: **O(V)**
5. The Minimum Spanning Tree efficiently connects all academic departments with minimum communication cost while maintaining complete network connectivity.

GitHub Link: https://github.com/vedapriya-vunnam/DSA2_CO3.git

Student Signature	**Faculty Signature**